

Predicting the Second Gift

Which first-time donors will give again, and which ones a small nonprofit should spend staff time on.

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DISCLOSURE, SAID FIRST

One of us founded GoGood Technologies, a donor-engagement platform whose customers are the stakeholder in this talk. That is why we chose the problem. Weigh what we say accordingly.

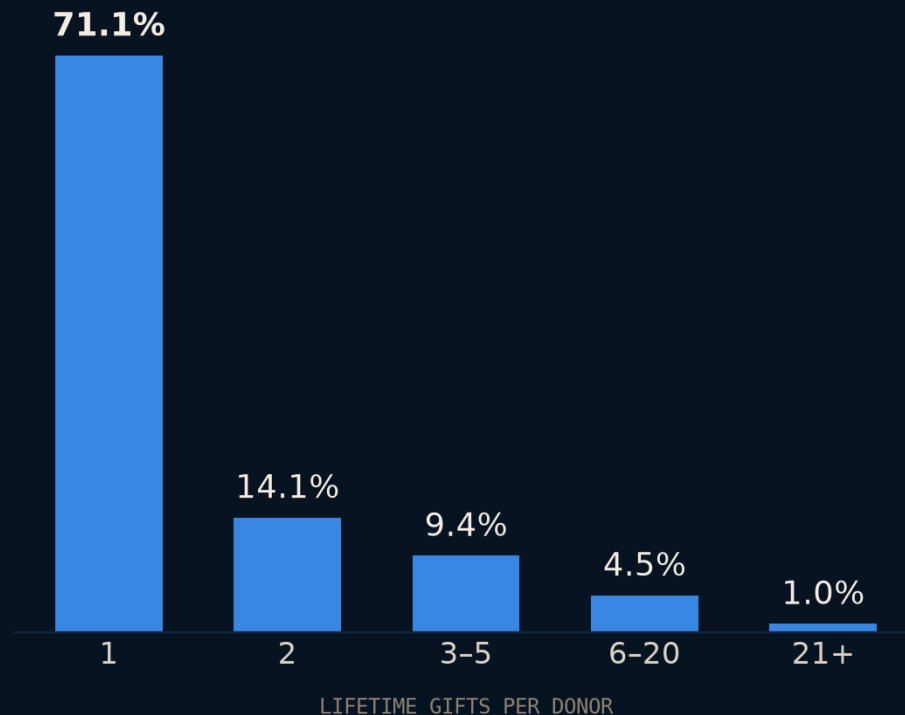
71%

of Donors Choose donors gave once and never again.

3.5 million donors, seventeen years. The second gift is where retention is won or lost, and small organizations have the least capacity to work it.

OUR STAKEHOLDER

A development lead at a ~\$500K nonprofit, no data staff. Each month she can personally follow up with a fraction of last month's new donors. She builds that list by hand, from recency and gift size.



Four decisions turn eleven million rows into one label. Each could have gone the other way.

11.4M

DONATIONS

3.28M

LABELLED DONORS

15.8%

REPEAT WITHIN 12 MO

ICPSR 37898, 2002–2019. Train through 2016, hold out 2017–18: a random split would let the model see the future.

1 Same-month repeats don't count

One checkout can fund several classrooms. Two gifts in month one are one event, not a return. She cannot steward back someone who never left.

2 Twelve whole months

M+1 to M+12. Dates are month-level, so no finer definition exists and none is claimed.

3 Unclosed windows: dropped, not zeroed

A 2019 donor has not had a year to come back. Labelling them 0 teaches the model that recent donors never return.

4 Refunds are not gifts

The codebook lists a minimum amount of –\$15. A reversal cannot open a cohort or count as a return. 151 rows, excluded.

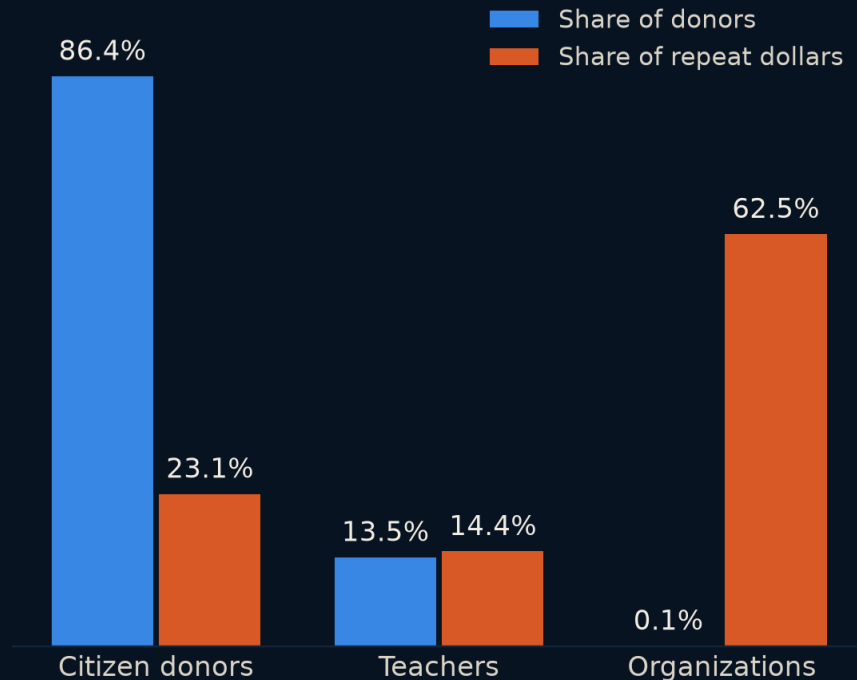
708 organizations hold 62% of every repeat dollar.

Our first result found 80% of all repeat dollars at 10% capacity with the dumbest possible rule: rank by gift size. It looked like a triumph.

It was measuring corporate matching programs. The largest made 66,348 donations in its twelve-month window. Any ranker wins the pooled contest by finding them — and she already knows who they are.

THE DECISION

Score citizen donors only — 86% of people, the ones she actually calls. One line in config, reversible, both numbers kept.



What predicts a return

Gift size finds dollars; [feature] finds people. Here is what moved and what did not.

Structure to keep: one chart of importance or lift, one sentence per feature that mattered, one on what did not. Text is masked by ICPSR — no embeddings. Tie back to slide 4.

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Feature importance or lift table — one chart

At her capacity, dollars identified per contact

Citizen donors · top 10% of each month's new donors · 2017–18 holdout · value identified, never “caused”

RANKING	PRECISION	RECALL	VALUE FOUND	\$ / CONTACT
Our model — $P(\text{return}) \times E[\text{amount}]$	22.2%	17.6%	58.3%	\$120
First gift size — the honest baseline	19.7%	15.7%	56.4%	\$116
First-month gift count	20.4%	16.2%	41.4%	\$85
Random	12.7%	10.1%	9.1%	\$19
Contact everyone	12.6%	100%	100%	\$21

Paired bootstrap on the same donors: $+\$3.80 \pm 0.51$ per contact over gift size, about 7 standard errors. Real, and small — gift size already carries most of what predicts dollars. The baseline is fair: on a first-gift cohort, RFM collapses to gift size, and it is what she does by hand.

How we know it works — and where it does not

It is calibrated here, it is not calibrated there, and it is weakest on exactly the donors she cares about most.

Calibration curve on the holdout. Error analysis by cohort year — does 2018 behave like 2015? — and by first-gift size band, where most of her donors live.

DROP HERE
Calibration curve

DROP HERE
*Error by cohort year / gift
band*

What she does on Monday

3,013

CALLS · DEC 2018

\$25

PER CALL ·
PLACEHOLDER

+\$7.75M

NET · 2017-18

\$0

TO RUN MONTHLY

Calling everyone loses \$3.6M at \$25 a contact. Calling the model's top 10% nets \$7.75M over the two holdout years — \$312K more than her gift-size rule on the same number of calls. Her cost per contact sets how deep to go: at \$5, half of new donors clear break-even; at \$25, 6.5%.

COST AT PRODUCTION SCALE

One monthly batch job on a laptop: about ten minutes, no API calls, no vendor. The only real cost is her time on the phone.

What we did not find, and what we are not claiming

1

No causal claim

Nobody was randomly assigned to be contacted. We rank by predicted future value and say “identified”, never “caused”.

2

Thank-you packet: dropped

The codebook gives no timing. The flag may record something that happened after the second gift. Albert pre-approved dropping it.

3

Transfer is a hypothesis

DonorsChoose donors are marketplace donors; small-nonprofit donors are relational. Behavioural features likely transfer; platform-specific ones likely do not.

Rank each month's first-time donors by predicted second-gift value, call the top one in ten, and expect to reach 58 percent of next year's repeat giving.